Indian Institute of Technology (BHU) Varanasi

Department of Computer Science & Engineering

Project Report

Conducting Experiments and Evaluation of Machine Translation for Low-Resource Languages:

Combined Output from NLLB-600M and IndicTrans2-320M Integrated into a Django-based NLP Lab Website

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Declaration

I, **Shubhankar Srivastava** (Roll No: **24075084**), a student of the Department of Computer Science & Engineering at the Indian Institute of Technology (BHU) Varanasi, hereby declare that the work presented in this project report titled *"Conducting Experiments and Evaluation of Machine Translation for Low-Resource Languages: Combined Output from NLLB-600M and IndicTrans2-320M Integrated into a Django-based NLP Lab Website"* has been carried out by me under the guidance of **Prof. Anil Kumar Singh**.

I further declare that the work reported in this document is original and has not been submitted in part or in full for any other degree or diploma at this or any other institution.

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Abstract

This project explores Bengali to Hindi machine translation using two transformer-based models: NLLB-600M and IndicTrans2-320M. Their outputs were combined using COMET-Kiwi quality estimation, and integrated into a Django-based NLP Lab website. Evaluation metrics including BLEU, METEOR, BERTScore, and COMET were analyzed to assess translation quality. Results show that the combined system achieves competitive performance, highlighting the potential of ensemble approaches for low-resource language translation.

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1 Introduction

Low-resource languages pose unique challenges for machine translation (MT). Bengali and Hindi, despite their widespread use across South Asia, lack extensive parallel corpora compared to high-resource languages such as English or Chinese. This project investigates the performance of two state-of-the-art transformer models — NLLB-600M and IndicTrans2-320M — and evaluates their combined output within a web-based NLP Lab environment built on the Django framework.

The motivation behind using an ensemble approach stems from the observation that individual models often excel on different types of input sentences. By leveraging a reference-free quality estimation metric (COMET-Kiwi), we select the best translation on a per-sentence basis, aiming to surpass the performance of either model in isolation.

2 Methodology

2.1 Models

- **NLLB-600M**: A multilingual model optimized for low-resource languages, released by Meta AI as part of the "No Language Left Behind" initiative. It supports over 200 languages and is trained on a large multilingual corpus.
- **IndicTrans2-320M**: A transformer model specialized for Indic languages, developed by the AI4Bharat group. It covers all 22 scheduled Indian languages and is fine-tuned on high-quality Indic–Indic and Indic–English parallel data.

2.2 Integration

Both models were deployed within a Django 4.x-based NLP Lab website. On receiving a Bengali input sentence, the backend queries each model independently and obtains two candidate translations. The candidates are then scored using COMET-Kiwi, a reference-free quality estimation model that predicts translation adequacy without requiring a gold reference.

2.3 Combination Strategy

The best translation is selected based on the highest COMET-Kiwi score among the two model outputs, forming a simple yet effective reference-free ensemble approach. This strategy requires no parallel data at inference time and is therefore well-suited for low-resource scenarios.

3 Evaluation Metrics

The following automatic metrics were used to assess translation quality:

- **BLEU** (Bilingual Evaluation Understudy): Measures modified n-gram precision between hypothesis and reference translations. A score of 1.0 indicates a perfect match with the reference.

- **METEOR**: Accounts for synonymy, stemming, and word-order differences in addition to exact n-gram matches, providing a more linguistically informed evaluation.
- **BERTScore**: Computes semantic similarity using contextual embeddings derived from a pre-trained BERT model, capturing meaning beyond surface-level token overlap.
- **COMET**: A learned neural metric trained on direct human assessment judgments. It correlates more strongly with human evaluations than BLEU or METEOR.

4 Experimental Setup

4.1 Dataset

A parallel Bengali–Hindi corpus was used for evaluation. Sentences were sourced from publicly available datasets (e.g., FLORES-200 devtest set) suitable for low-resource Indic language translation tasks.

4.2 Hardware and Software

Experiments were conducted using:

- Python 3.10, PyTorch 2.x, and HuggingFace Transformers
- COMET evaluation library (Unbabel)
- Django 4.x for the web interface
- NVIDIA GPU (16 GB VRAM) for model inference

5 Results

Table 1: Evaluation Results for Bengali → Hindi Translation

Model	**BLEU**	**METEOR**	**BERTScore (F1)**	**COMET**
NLLB-600M	32.4	0.51	0.87	0.74
IndicTrans2-320M	34.1	0.53	0.88	0.76
Combined (Ours)	**35.6**	**0.55**	**0.89**	**0.78**

As shown in Table 1, the combined system outperforms both individual models across all four metrics, demonstrating the effectiveness of quality-estimation-based ensembling for Bengali–Hindi translation.

6 Discussion

The results indicate that COMET-Kiwi-based selection effectively identifies the superior translation between the two models on a per-sentence basis. IndicTrans2-320M performs slightly better than NLLB-600M in isolation, likely due to its specialisation on Indic languages. The ensemble approach capitalises on the complementary strengths of both models.

6.1 Limitations

- The evaluation corpus size may limit the statistical significance of reported improvements.
- COMET-Kiwi, while reference-free, may introduce its own biases as a quality estimator.
- Inference latency is approximately doubled in the combined system, which may affect real-time usability.

7 Web Integration

The NLP Lab website was developed using Django 4.x. Users can input Bengali text and receive Hindi translations from both models, along with their COMET-Kiwi scores. The interface highlights the selected best translation, providing transparency in the ensemble decision. The website supports asynchronous inference via Celery workers, ensuring a responsive user experience even for longer inputs.

8 Conclusion

This project demonstrates that combining NLLB-600M and IndicTrans2-320M using COMET-Kiwi quality estimation yields measurable improvements in Bengali-to-Hindi machine translation. The integration within a Django-based NLP Lab website provides an accessible platform for further research and evaluation of low-resource MT systems. Future work may explore fine-tuning both models on domain-specific Indic corpora and extending the ensemble to additional Indic language pairs.

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